



Instructor: Date:

Learner: Regional Branch:

Program: Center:

CONTENT

- Let's Start! • Let's Explore! • Let's Practice! • Extension Activity • Target Words • Learning Points



Let's Start!

Is it valid or not valid? That's the question!



Learning Outcome:

Designs the data structure based on the database structures and its characteristics.

Validate database using the proper IT tools.

Can Do Statements:

- I can talk about where data validation was performed in a system.
- I can describe the different data validation techniques used in the system.

Ask your partner:

- Do you think that the same validation technique works for all sorts of systems? Why? Why not?
- Do you believe that we can create validation-free systems? Why? Why not?





Let's Explore!

Giving accounts* to my boss...

*"Giving accounts" is an expression we use to say that we need to describe our actions to our superiors.



Instructions: Read the text below.

- Thomas:** Last week I worked from home so I'd better report to my boss what I did. Here she comes. Boss, I think you might want to check what I have done since last Monday when I started teleworking.
- Susan:** Sure, show me that.
- Thomas:** I've studied the parts where data validation ought to be performed in our current project. Therefore, I started with the user interactivity screens and forms. So, I identified all the different parts of the application that required interaction from the users, as those who were more susceptible to suffer from human mistakes. Then, I continued with the file manipulation routines, which were basically the parts in which reading and writing operations from and to the file were involved.
- Susan:** Correct. Remember that you ought to include the import and export routines too in which customers might want to save data in different file formats for other types of systems. Now, what techniques did you use?
- Thomas:** I decided to carry out two different techniques I have mastered throughout my work here. First, I used the range validation technique, which I applied to the numbers and dates in order to confirm that the registered values were within the specific minimum and maximum ranges. Then, I proceeded with the lookup validation in which I contrasted the data with other possible values and data, just to make sure that they were valid.
- Susan:** You'd better use the masked input validation, right? It is important that you pay attention to those values in which specific input patterns have to be included. Plus, you ought to make sure that telephone numbers and dates were correctly arrayed with their respective patterns. If you have any questions, you may want to call me at the office.

**Let's Practice!**

Susan can determine those concepts after Thomas explanation.

Instructions: Select the correct category in the table. There is only one answer per category.

	telework	interactivity	susceptible	technique
A way someone performs a task with special skills				
easily affected or damaged by something				
electronic communication system which involve user orders and system reaction				
working from home over the computer				
numeric quantity assigned to something base on a measure unit				
to show or portray				

**Let's Practice!**

That was the system which I performed.

Instructions: Read the sentences. Write the missing word(s) in the blank. Use: continued, contrasted, have mastered, identified, have used

1. I started with the user interactivity screens and forms, so I _____ all the different parts of the application.
2. I _____ with the file manipulation routines which were basically the parts in which reading and writing operations from and to the file were involved.
3. I decided to carry out two different techniques I _____ throughout my work here.
4. I _____ the range validation technique which I've applied to the numbers and dates in order to confirm that the registered values were within the specific minimum and maximum ranges.
5. I proceeded with the lookup validation in which I _____ the data with other possible values and data just to make sure that they were valid.



Extension Activity

Let's write about the validation techniques and your own style.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for more information about data validation techniques.
- Write notes about how you apply them into a system development.
- Make a presentation in order to explain the different techniques.
- Share it with your classmates and answer all his questions.

Self-Assessment

Yes No

- I look for further information about data validation techniques.
- I express information through a poster.
- I share information with partners.

☐	☐
☐	☐
☐	☐



Target Words

- telework** (noun): working from home over the computer
e.g. Boss, I think you may want to check what I have done since last Monday when I started teleworking.
- current** (adjective): actual, in the present moment
e.g. I've studied the parts where data validation ought to be performed in our current project.
- susceptible** (adjective): easy to influence or occur
e.g. I identified all the different parts of the application that required interaction from the users, as those who were more susceptible to suffer from human mistakes.
- master** (verb): to reach a high level of knowledge and skill
e.g. I decided to carry out two different techniques I have mastered throughout my work here.
- range** (noun): a group of values
e.g. First, I used the range validation technique which I applied to the numbers and dates in order to confirm that the registered values were within the specific minimum and maximum ranges.
- array** (verb): to show or portray
e.g. Plus, you ought to make sure that telephone numbers and dates were correctly arrayed with their respective patterns.



Learning Points

Present Perfect Versus Past Simple- Affirmative Form

The **Present Perfect** as well as the **Past Simple** refer to two different moments in time.

PRESENT PERFECT

You use the **present perfect** tense to talk about an action or situation that started in the past, but continues up to the present moment (unfinished actions). Remember that the **present perfect** tense needs the auxiliary verb **have** (I/you/we/they) or **has** (he/she/it) plus the past participle form of the verb.

Example:

*Check what I **have done** since last Monday.*

PAST SIMPLE

You use the **past simple** tense to talk about an action or situation that took place in the past (finished actions). Remember that depending on the type of the verb, regular or irregular, the verb will change differently to its **simple past** form.

Example:

*Last week I **worked** from home so I'd better report to my boss what I **did**.*

Examples

- *Boss, I think you may want to check what I have done since last Monday when I started teleworking.*
- *I decided to carry out two different techniques I have mastered throughout my work here.*





Learning Points

Data Validation Techniques

There are different techniques that help us validate data in big projects and avoid errors as much as possible.

This technique applies only to dates and numbers. A test is performed in order to confirm that the registered values are in specific minimum and maximum ranges.

LOOKUP VALIDATION

This validation technique is performed when the registered data has to be contrasted with other possible values or data. A comparison is performed in order to state whether the entries are valid or not.

MASKED INPUT VALIDATION

There are some data or values that require a specific input pattern to be registered in the databases. Therefore, this technique, also known as Filtered Input Validation, only allows registering specific data that follows a pre-established pattern or data type. This is useful when pieces of information such as dates, telephone numbers, among others.